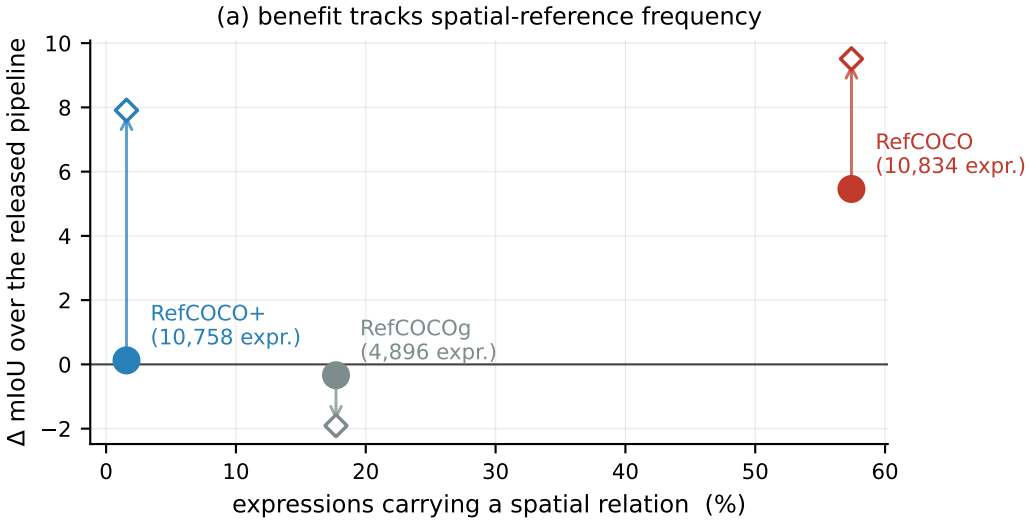
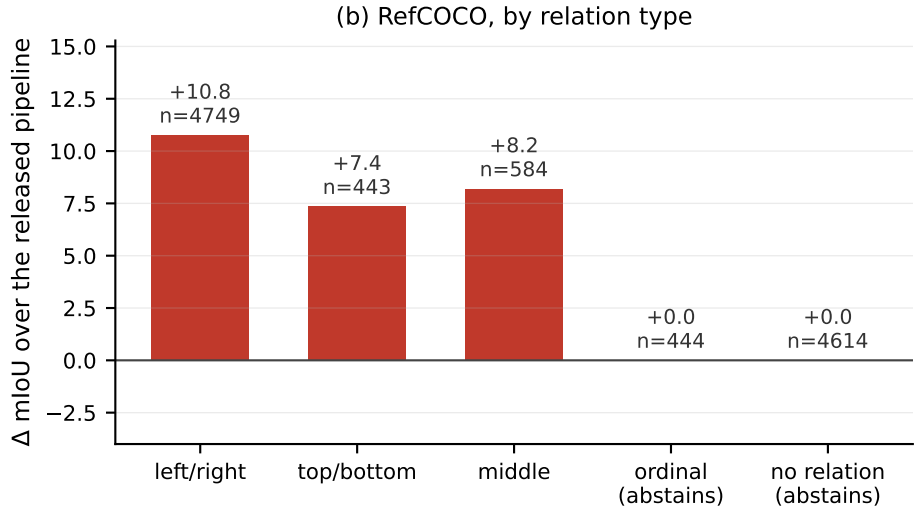

(a) benefit tracks spatial-reference frequency
Δ mIoU over the released pipeline
10
8
6
4
2
0
−2
0
10
20
30
40
50
60
expressions carrying a spatial relation (%)
RefCOCO+
(10,758 expr.)
RefCOCOg
(4,896 expr.)
RefCOCO
(10,834 expr.)
filled: all expressions · hollow diamond: spatial stratum only


(b) RefCOCO, by relation type
Δ mIoU over the released pipeline
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
+10.8
n=4749
+7.4
n=443
+8.2
n=584
+0.0
n=444
+0.0
n=4614
left/right
top/bottom
middle
ordinal
(abstains)
no relation
(abstains)